

Student Evaluation MySQL Flat Table Specification

Formal Technical Documentation — MySQL Dedicated Flat Tables & 15-Char Register Number Primary Key

Database Engine: MySQL Server 8.0+	Primary Key: id (VARCHAR(50))	Conflict Strategy: ON DUPLICATE KEY UPDATE
Tables: Dedicated Per-Semester	Package: student_migrator	File Formats: Excel (.xlsx) & CSV (.csv)

This document details the MySQL database flat tables (student_track_evaluation_sem2, student_track_evaluation_sem3, student_track_evaluation_sem4) using the 15-character student register number (id) as the **PRIMARY KEY**.

1. Database Table Architecture

Instead of storing surrogate integer keys (student_id), the database uses id (**15-character student register number e.g. RA2411029010002**) **directly as the PRIMARY KEY**:

- student_track_evaluation_sem2 — Semester 2 Flat Table (id VARCHAR(50) PRIMARY KEY)
- student_track_evaluation_sem3 — Semester 3 Flat Table (id VARCHAR(50) PRIMARY KEY)
- student_track_evaluation_sem4 — Semester 4 Flat Table (id VARCHAR(50) PRIMARY KEY)

2. Semester 2 Flat Table DDL (`student\_track\_evaluation\_sem2`)

```
CREATE TABLE IF NOT EXISTS student_track_evaluation_sem2 (  
  id VARCHAR(50) PRIMARY KEY,  
  sn INT,  
  full_name VARCHAR(255),  
  college_email_id VARCHAR(255),  
  assignment_submission VARCHAR(255),  
  engagement_status VARCHAR(255),  
  
  core_level TEXT,  
  core_class_practice VARCHAR(255),  
  core_assignment VARCHAR(255),  
  core_tech_input_t1 DECIMAL(5,2),  
  core_tech_input_t2 DECIMAL(5,2),  
  core_tech_input_t3 DECIMAL(5,2),  
  core_tech_input_t4 DECIMAL(5,2),  
  core_learn_input_l1 DECIMAL(5,2),  
  core_learn_input_l2 DECIMAL(5,2),  
  core_learn_input_l3 DECIMAL(5,2),  
  core_learn_input_l4 DECIMAL(5,2),  
  core_comm_input_c1 DECIMAL(5,2),  
  core_comm_input_c2 DECIMAL(5,2),  
  core_comm_input_c3 DECIMAL(5,2),  
  core_comm_input_c4 DECIMAL(5,2),  
  core_tech_score_t1 DECIMAL(5,2),  
  core_tech_score_t2 DECIMAL(5,2),  
  core_tech_score_t3 DECIMAL(5,2),  
  core_tech_score_t4 DECIMAL(5,2),  
  core_learn_score_l1 DECIMAL(5,2),  
  core_learn_score_l2 DECIMAL(5,2),  
  core_learn_score_l3 DECIMAL(5,2),  
  core_learn_score_l4 DECIMAL(5,2),  
  core_learn_score_combined DECIMAL(5,2),
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core_level_computed TEXT,
core_comm_score_c1 DECIMAL(5,2),
core_comm_score_c2 DECIMAL(5,2),
core_comm_score_c3 DECIMAL(5,2),
core_comm_score_c4 DECIMAL(5,2),
core_summary_t DECIMAL(5,2),
core_summary_l DECIMAL(5,2),
core_summary_c DECIMAL(5,2),
core_total_score DECIMAL(5,2),
core_help_status VARCHAR(255),
core_coding_level TEXT,
core_review_comments TEXT,

logical_level TEXT,
logical_class_practice VARCHAR(255),
logical_assignment VARCHAR(255),
logical_tech_input_t1 DECIMAL(5,2),
logical_tech_input_t2 DECIMAL(5,2),
logical_tech_input_t3 DECIMAL(5,2),
logical_tech_input_t4 DECIMAL(5,2),
logical_learn_input_l1 DECIMAL(5,2),
logical_learn_input_l2 DECIMAL(5,2),
logical_learn_input_l3 DECIMAL(5,2),
logical_learn_input_l4 DECIMAL(5,2),
logical_comm_input_c1 DECIMAL(5,2),
logical_comm_input_c2 DECIMAL(5,2),
logical_comm_input_c3 DECIMAL(5,2),
logical_comm_input_c4 DECIMAL(5,2),
logical_tech_score_t1 DECIMAL(5,2),
logical_tech_score_t2 DECIMAL(5,2),
logical_tech_score_t3 DECIMAL(5,2),
logical_tech_score_t4 DECIMAL(5,2),
logical_learn_score_l1 DECIMAL(5,2),
logical_learn_score_l2 DECIMAL(5,2),
logical_learn_score_l3 DECIMAL(5,2),
logical_learn_score_l4 DECIMAL(5,2),
logical_learn_score_combined DECIMAL(5,2),
logical_level_computed TEXT,
logical_comm_score_c1 DECIMAL(5,2),
logical_comm_score_c2 DECIMAL(5,2),
logical_comm_score_c3 DECIMAL(5,2),
logical_comm_score_c4 DECIMAL(5,2),
logical_summary_t DECIMAL(5,2),
logical_summary_l DECIMAL(5,2),
logical_summary_c DECIMAL(5,2),
logical_total_score DECIMAL(5,2),
logical_help_status VARCHAR(255),
logical_coding_level TEXT,
logical_review_comments TEXT,

oop_level TEXT,
oop_class_practice VARCHAR(255),
oop_assignment VARCHAR(255),
oop_tech_input_t1 DECIMAL(5,2),
oop_tech_input_t2 DECIMAL(5,2),
oop_tech_input_t3 DECIMAL(5,2),
oop_tech_input_t4 DECIMAL(5,2),
oop_learn_input_l1 DECIMAL(5,2),
oop_learn_input_l2 DECIMAL(5,2),
oop_learn_input_l3 DECIMAL(5,2),
oop_learn_input_l4 DECIMAL(5,2),
oop_comm_input_c1 DECIMAL(5,2),
oop_comm_input_c2 DECIMAL(5,2),
oop_comm_input_c3 DECIMAL(5,2),

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    oop_comm_input_c4 DECIMAL(5,2),
    oop_tech_score_t1 DECIMAL(5,2),
    oop_tech_score_t2 DECIMAL(5,2),
    oop_tech_score_t3 DECIMAL(5,2),
    oop_tech_score_t4 DECIMAL(5,2),
    oop_learn_score_l1 DECIMAL(5,2),
    oop_learn_score_l2 DECIMAL(5,2),
    oop_learn_score_l3 DECIMAL(5,2),
    oop_learn_score_l4 DECIMAL(5,2),
    oop_learn_score_combined DECIMAL(5,2),
    oop_level_computed TEXT,
    oop_comm_score_c1 DECIMAL(5,2),
    oop_comm_score_c2 DECIMAL(5,2),
    oop_comm_score_c3 DECIMAL(5,2),
    oop_comm_score_c4 DECIMAL(5,2),
    oop_summary_t DECIMAL(5,2),
    oop_summary_l DECIMAL(5,2),
    oop_summary_c DECIMAL(5,2),
    oop_total_score DECIMAL(5,2),
    oop_help_status VARCHAR(255),
    oop_coding_level TEXT,
    oop_review_comments TEXT,

    core_final VARCHAR(255),
    logical_final VARCHAR(255),
    oop_final VARCHAR(255),
    final_avg_track VARCHAR(255)
);

```

3. Semester 3 Flat Table DDL (`student\_track\_evaluation\_sem3`)

Subjects: string_* (String Fundamentals...), oop_* (Core OOP Principles), adv_oop_* (Advanced OOP & Data Structures).

```

CREATE TABLE IF NOT EXISTS student_track_evaluation_sem3 (
    id VARCHAR(50) PRIMARY KEY,
    sn INT,
    full_name VARCHAR(255),
    college_email_id VARCHAR(255),
    assignment_submission VARCHAR(255),
    engagement_status VARCHAR(255),

    string_level TEXT,
    string_class_practice VARCHAR(255),
    string_assignment VARCHAR(255),
    string_tech_input_t1 DECIMAL(5,2),
    string_tech_input_t2 DECIMAL(5,2),
    string_tech_input_t3 DECIMAL(5,2),
    string_tech_input_t4 DECIMAL(5,2),
    string_learn_input_l1 DECIMAL(5,2),
    string_learn_input_l2 DECIMAL(5,2),
    string_learn_input_l3 DECIMAL(5,2),
    string_learn_input_l4 DECIMAL(5,2),
    string_comm_input_c1 DECIMAL(5,2),
    string_comm_input_c2 DECIMAL(5,2),
    string_comm_input_c3 DECIMAL(5,2),
    string_comm_input_c4 DECIMAL(5,2),
    string_tech_score_t1 DECIMAL(5,2),
    string_tech_score_t2 DECIMAL(5,2),
    string_tech_score_t3 DECIMAL(5,2),
    string_tech_score_t4 DECIMAL(5,2),
    string_learn_score_l1 DECIMAL(5,2),
    string_learn_score_l2 DECIMAL(5,2),

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string_learn_score_l3 DECIMAL(5,2),
string_learn_score_l4 DECIMAL(5,2),
string_learn_score_combined DECIMAL(5,2),
string_level_computed TEXT,
string_comm_score_c1 DECIMAL(5,2),
string_comm_score_c2 DECIMAL(5,2),
string_comm_score_c3 DECIMAL(5,2),
string_comm_score_c4 DECIMAL(5,2),
string_summary_t DECIMAL(5,2),
string_summary_l DECIMAL(5,2),
string_summary_c DECIMAL(5,2),
string_total_score DECIMAL(5,2),
string_help_status VARCHAR(255),
string_coding_level TEXT,
string_review_comments TEXT,

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oop_level TEXT,
oop_class_practice VARCHAR(255),
oop_assignment VARCHAR(255),
oop_tech_input_t1 DECIMAL(5,2),
oop_tech_input_t2 DECIMAL(5,2),
oop_tech_input_t3 DECIMAL(5,2),
oop_tech_input_t4 DECIMAL(5,2),
oop_learn_input_l1 DECIMAL(5,2),
oop_learn_input_l2 DECIMAL(5,2),
oop_learn_input_l3 DECIMAL(5,2),
oop_learn_input_l4 DECIMAL(5,2),
oop_comm_input_c1 DECIMAL(5,2),
oop_comm_input_c2 DECIMAL(5,2),
oop_comm_input_c3 DECIMAL(5,2),
oop_comm_input_c4 DECIMAL(5,2),
oop_tech_score_t1 DECIMAL(5,2),
oop_tech_score_t2 DECIMAL(5,2),
oop_tech_score_t3 DECIMAL(5,2),
oop_tech_score_t4 DECIMAL(5,2),
oop_learn_score_l1 DECIMAL(5,2),
oop_learn_score_l2 DECIMAL(5,2),
oop_learn_score_l3 DECIMAL(5,2),
oop_learn_score_l4 DECIMAL(5,2),
oop_learn_score_combined DECIMAL(5,2),
oop_level_computed TEXT,
oop_comm_score_c1 DECIMAL(5,2),
oop_comm_score_c2 DECIMAL(5,2),
oop_comm_score_c3 DECIMAL(5,2),
oop_comm_score_c4 DECIMAL(5,2),
oop_summary_t DECIMAL(5,2),
oop_summary_l DECIMAL(5,2),
oop_summary_c DECIMAL(5,2),
oop_total_score DECIMAL(5,2),
oop_help_status VARCHAR(255),
oop_coding_level TEXT,
oop_review_comments TEXT,

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adv_oop_level TEXT,
adv_oop_class_practice VARCHAR(255),
adv_oop_assignment VARCHAR(255),
adv_oop_tech_input_t1 DECIMAL(5,2),
adv_oop_tech_input_t2 DECIMAL(5,2),
adv_oop_tech_input_t3 DECIMAL(5,2),
adv_oop_tech_input_t4 DECIMAL(5,2),
adv_oop_learn_input_l1 DECIMAL(5,2),
adv_oop_learn_input_l2 DECIMAL(5,2),
adv_oop_learn_input_l3 DECIMAL(5,2),
adv_oop_learn_input_l4 DECIMAL(5,2),

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adv_oop_comm_input_c1 DECIMAL(5,2),
adv_oop_comm_input_c2 DECIMAL(5,2),
adv_oop_comm_input_c3 DECIMAL(5,2),
adv_oop_comm_input_c4 DECIMAL(5,2),
adv_oop_tech_score_t1 DECIMAL(5,2),
adv_oop_tech_score_t2 DECIMAL(5,2),
adv_oop_tech_score_t3 DECIMAL(5,2),
adv_oop_tech_score_t4 DECIMAL(5,2),
adv_oop_learn_score_l1 DECIMAL(5,2),
adv_oop_learn_score_l2 DECIMAL(5,2),
adv_oop_learn_score_l3 DECIMAL(5,2),
adv_oop_learn_score_l4 DECIMAL(5,2),
adv_oop_learn_score_combined DECIMAL(5,2),
adv_oop_level_computed TEXT,
adv_oop_comm_score_c1 DECIMAL(5,2),
adv_oop_comm_score_c2 DECIMAL(5,2),
adv_oop_comm_score_c3 DECIMAL(5,2),
adv_oop_comm_score_c4 DECIMAL(5,2),
adv_oop_summary_t DECIMAL(5,2),
adv_oop_summary_l DECIMAL(5,2),
adv_oop_summary_c DECIMAL(5,2),
adv_oop_total_score DECIMAL(5,2),
adv_oop_help_status VARCHAR(255),
adv_oop_coding_level TEXT,
adv_oop_review_comments TEXT,

string_final VARCHAR(255),
oop_final VARCHAR(255),
adv_oop_final VARCHAR(255),
final_avg_track VARCHAR(255)
);

```

4. Semester 4 Flat Table DDL (`student\_track\_evaluation\_sem4`)

Subjects: dsa_* (Data Structures & Algorithms), collections_* (Collections & Stream API), java_adv_* (Java Advanced).

```

CREATE TABLE IF NOT EXISTS student_track_evaluation_sem4 (
  id VARCHAR(50) PRIMARY KEY,
  sn INT,
  full_name VARCHAR(255),
  college_email_id VARCHAR(255),
  assignment_submission VARCHAR(255),
  engagement_status VARCHAR(255),

  dsa_level TEXT,
  dsa_class_practice VARCHAR(255),
  dsa_assignment VARCHAR(255),
  dsa_tech_input_t1 DECIMAL(5,2),
  dsa_tech_input_t2 DECIMAL(5,2),
  dsa_tech_input_t3 DECIMAL(5,2),
  dsa_tech_input_t4 DECIMAL(5,2),
  dsa_learn_input_l1 DECIMAL(5,2),
  dsa_learn_input_l2 DECIMAL(5,2),
  dsa_learn_input_l3 DECIMAL(5,2),
  dsa_learn_input_l4 DECIMAL(5,2),
  dsa_comm_input_c1 DECIMAL(5,2),
  dsa_comm_input_c2 DECIMAL(5,2),
  dsa_comm_input_c3 DECIMAL(5,2),
  dsa_comm_input_c4 DECIMAL(5,2),
  dsa_tech_score_t1 DECIMAL(5,2),
  dsa_tech_score_t2 DECIMAL(5,2),
  dsa_tech_score_t3 DECIMAL(5,2),

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dsa_tech_score_t4 DECIMAL(5,2),
dsa_learn_score_l1 DECIMAL(5,2),
dsa_learn_score_l2 DECIMAL(5,2),
dsa_learn_score_l3 DECIMAL(5,2),
dsa_learn_score_l4 DECIMAL(5,2),
dsa_learn_score_combined DECIMAL(5,2),
dsa_level_computed TEXT,
dsa_comm_score_c1 DECIMAL(5,2),
dsa_comm_score_c2 DECIMAL(5,2),
dsa_comm_score_c3 DECIMAL(5,2),
dsa_comm_score_c4 DECIMAL(5,2),
dsa_summary_t DECIMAL(5,2),
dsa_summary_l DECIMAL(5,2),
dsa_summary_c DECIMAL(5,2),
dsa_total_score DECIMAL(5,2),
dsa_help_status VARCHAR(255),
dsa_coding_level TEXT,
dsa_review_comments TEXT,

collections_level TEXT,
collections_class_practice VARCHAR(255),
collections_assignment VARCHAR(255),
collections_tech_input_t1 DECIMAL(5,2),
collections_tech_input_t2 DECIMAL(5,2),
collections_tech_input_t3 DECIMAL(5,2),
collections_tech_input_t4 DECIMAL(5,2),
collections_learn_input_l1 DECIMAL(5,2),
collections_learn_input_l2 DECIMAL(5,2),
collections_learn_input_l3 DECIMAL(5,2),
collections_learn_input_l4 DECIMAL(5,2),
collections_comm_input_c1 DECIMAL(5,2),
collections_comm_input_c2 DECIMAL(5,2),
collections_comm_input_c3 DECIMAL(5,2),
collections_comm_input_c4 DECIMAL(5,2),
collections_tech_score_t1 DECIMAL(5,2),
collections_tech_score_t2 DECIMAL(5,2),
collections_tech_score_t3 DECIMAL(5,2),
collections_tech_score_t4 DECIMAL(5,2),
collections_learn_score_l1 DECIMAL(5,2),
collections_learn_score_l2 DECIMAL(5,2),
collections_learn_score_l3 DECIMAL(5,2),
collections_learn_score_l4 DECIMAL(5,2),
collections_learn_score_combined DECIMAL(5,2),
collections_level_computed TEXT,
collections_comm_score_c1 DECIMAL(5,2),
collections_comm_score_c2 DECIMAL(5,2),
collections_comm_score_c3 DECIMAL(5,2),
collections_comm_score_c4 DECIMAL(5,2),
collections_summary_t DECIMAL(5,2),
collections_summary_l DECIMAL(5,2),
collections_summary_c DECIMAL(5,2),
collections_total_score DECIMAL(5,2),
collections_help_status VARCHAR(255),
collections_coding_level TEXT,
collections_review_comments TEXT,

java_adv_level TEXT,
java_adv_class_practice VARCHAR(255),
java_adv_assignment VARCHAR(255),
java_adv_tech_input_t1 DECIMAL(5,2),
java_adv_tech_input_t2 DECIMAL(5,2),
java_adv_tech_input_t3 DECIMAL(5,2),
java_adv_tech_input_t4 DECIMAL(5,2),
java_adv_learn_input_l1 DECIMAL(5,2),

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java_adv_learn_input_l2 DECIMAL(5,2),
java_adv_learn_input_l3 DECIMAL(5,2),
java_adv_learn_input_l4 DECIMAL(5,2),
java_adv_comm_input_c1 DECIMAL(5,2),
java_adv_comm_input_c2 DECIMAL(5,2),
java_adv_comm_input_c3 DECIMAL(5,2),
java_adv_comm_input_c4 DECIMAL(5,2),
java_adv_tech_score_t1 DECIMAL(5,2),
java_adv_tech_score_t2 DECIMAL(5,2),
java_adv_tech_score_t3 DECIMAL(5,2),
java_adv_tech_score_t4 DECIMAL(5,2),
java_adv_learn_score_l1 DECIMAL(5,2),
java_adv_learn_score_l2 DECIMAL(5,2),
java_adv_learn_score_l3 DECIMAL(5,2),
java_adv_learn_score_l4 DECIMAL(5,2),
java_adv_learn_score_combined DECIMAL(5,2),
java_adv_level_computed TEXT,
java_adv_comm_score_c1 DECIMAL(5,2),
java_adv_comm_score_c2 DECIMAL(5,2),
java_adv_comm_score_c3 DECIMAL(5,2),
java_adv_comm_score_c4 DECIMAL(5,2),
java_adv_summary_t DECIMAL(5,2),
java_adv_summary_l DECIMAL(5,2),
java_adv_summary_c DECIMAL(5,2),
java_adv_total_score DECIMAL(5,2),
java_adv_help_status VARCHAR(255),
java_adv_coding_level TEXT,
java_adv_review_comments TEXT,

dsa_final VARCHAR(255),
collections_final VARCHAR(255),
java_adv_final VARCHAR(255),
final_avg_track VARCHAR(255)
);
```

5. Logical Unique Key & Upsert Rules

- **Primary Key:** `id` (student 15-character register number e.g. RA2411029010002).
- **Behavior:** Re-running a migration updates existing records (`ON DUPLICATE KEY UPDATE`) in-place based on `id` without creating duplicate rows.